

Detecting Toxic Flow

Álvaro Cartea^{a,b}, Gerardo Duran-Martin^{b,c}, Leandro Sánchez-Betancourt^{a,b,d}

^a*Mathematical Institute, University of Oxford, Oxford, UK*

^b*Oxford-Man Institute of Quantitative Finance, Oxford, UK*

^c*School of Mathematical Sciences, Queen Mary University, London, UK*

^d*LMAX Exchange, London, UK*

Abstract

This paper develops a framework to predict toxic trades that a broker receives from her clients. Toxic trades are predicted with a novel online Bayesian method which we call the *projection-based unification of last-layer and subspace estimation* (PULSE). PULSE is a fast and statistically-efficient online procedure to train a Bayesian neural network sequentially. We employ a proprietary dataset of foreign exchange transactions to test our methodology. PULSE outperforms standard machine learning and statistical methods when predicting if a trade will be toxic; the benchmark methods are logistic regression, random forests, and a recursively-updated maximum-likelihood estimator. We devise a strategy for the broker who uses toxicity predictions to internalise or to externalise each trade received from her clients. Our methodology can be implemented in real-time because it takes less than one millisecond to update parameters and make a prediction. Compared with the benchmarks, PULSE attains the highest PnL and the largest avoided loss for the horizons we consider.

1. Introduction

Liquidity providers are key to well-functioning financial markets. In foreign exchange (FX), as in other asset classes, broker-client relationships are ubiquitous. The broker streams bid and ask quotes to her clients and the clients decide when to trade on these quotes, so the broker bears the risk of adverse selection when trading with better informed clients. These risks are borne by both liquidity providers who stream quotes to individual parties and by market participants who provide liquidity in the books of electronic exchanges. However, in contrast to electronic order books in which trading is anonymous for all participants (e.g., in Nasdaq, LSE, Euronext), in broker-client relationships the broker knows which client executed the order. This privileged information can be used by the broker to classify flow, i.e., toxic or benign, and to devise strategies that mitigate adverse selection costs.

In the literature, models generally classify traders as informed or uninformed; see e.g., [Bagehot \(1971\)](#), [Copeland and Galai \(1983\)](#), [Grossman and Stiglitz \(1980\)](#), [Amihud and Mendelson \(1980\)](#), [Kyle \(1989\)](#), [Kyle \(1985\)](#), and [Glosten and Milgrom \(1985\)](#). In equity markets, many studies focus on informed flow (i.e., asymmetry of information) across various traded stocks, see e.g., [Easley et al. \(1996\)](#) who study the probability of informed trading at the stock level, while our study focuses on

*We thank Andrew Stewart, Alistair Sturgiss, Fayçal Drissi, Patrick Chang, Álvaro Arroyo, Sergio Calvo Ordoñez, and participants at the Oxford Victoria Seminar for comments. ChatGPT suggested the name PULSE for our algorithm.

each trade because we have trader identification. In FX markets, [Butz and Oomen \(2019\)](#) develop a model for internalisation in FX markets, and [Oomen \(2017\)](#) studies execution in an FX aggregator and the market impact of internalisation-externalisation strategies. Overall, studies of toxic flow and information asymmetry do not make predictions of toxicity at the trade level. To the best of our knowledge, ours is the first paper in the literature to use FX data with trader identification to predict the toxicity of each trade.

In our work, a trade is toxic if a client can unwind the trade within a given time window and make a profit (i.e., a loss for the broker). Toxic trades are not necessarily informed, nor informed trades are necessarily toxic. An uninformed client can execute a trade that becomes toxic for the broker because of the random fluctuations of exchange rates. Ultimately, the broker’s objective is to avoid holding loss-leading trades in her books, so it is more effective to focus on market features and each trade the broker fills, rather than on whether a particular client is classified as informed or uninformed. For simplicity, theoretical models in the literature assume traders are informed or uninformed, while in practice not all trades sent by one particular client are motivated by superior information.¹

The main contributions of our paper are as follows. We predict the toxicity of each incoming trade with machine learning and statistical methods, such as logistic regression, random forests, a recursively updated maximum-likelihood estimator, and develop a novel algorithm that uses a neural network (NNet) and that is able to update the model parameters sequentially and efficiently. We call this new method PULSE, which stands for projection-based unification of last-layer and subspace estimation.

Our new method employs a neural network to compute the probability that a trade will be toxic. After revealing if a trade was toxic, PULSE uses the last observed features to update the parameters of the NNet. To estimate parameters efficiently at each timestep, we follow three steps. One, we split the last layer from the feature-transformation layers of a NNet. Two, we project the parameters of the feature-transformation layers onto an affine subspace. Three, we devise a recursive formula to estimate a posterior distribution over the projected feature-transformation parameters and last-layer parameters. Specifically, we extend the subspace neural network model (subspace NNets) of [Duran-Martin et al. \(2022\)](#) to classification tasks. We also use the exponential-family extended Kalman filter (expfam EKF) method of [Ollivier \(2017\)](#) and we follow the ideas of the recursive variational Gaussian approximation (R-VGA) results of [Lambert et al. \(2021\)](#) to obtain the update equations in PULSE. Finally, we impose a prior independence between the hidden layers of the neural network and the output layer, extending the work in the last-layer Bayesian neural networks literature (last-layer BNNs). Figure 1 shows the relationship of PULSE to previous methods.

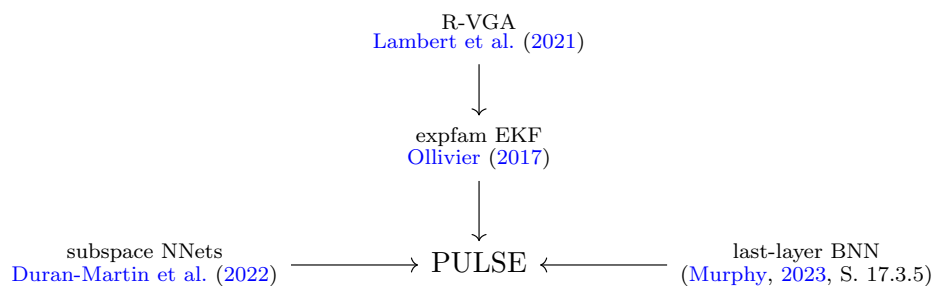


Figure 1: Relationship of PULSE to other models.

¹[Cartea and Sánchez-Betancourt \(2022\)](#) show how the broker optimally provides liquidity to clients who have been previously classified as informed or uninformed.

To evaluate the predictive performance of our model and the efficacy of the broker's strategy, we use a proprietary dataset of FX transactions from 28 June 2022 to 21 October 2022. Initially, the models are trained with data between 28 June and 31 July, and the remainder of the data (1 August to 21 October) is used to deploy the strategy, i.e., use predictions of toxicity for each trade to inform the internalisation-externalisation strategy we develop. During the deploy phase, PULSE and the maximum-likelihood estimator models continue to learn, while the models based on logistic regression and random forests are not updated. For a given toxicity horizon and a cutoff probability the (internalisation-externalisation) strategy internalises the trade if the probability that the trade is toxic is less than or equal to the cutoff probability, otherwise it externalises the trade. We compute the PnL of all trades that the broker internalised and the avoided loss due to externalised trades. We find that PULSE delivers the best combination of PnL and avoided loss across all toxicity horizons we consider in this paper.

Finally, it is more advantageous to have a universal model than to have one model per trader. That is, we obtain higher accuracies when we train one model for all traders than when we train one model per trader; higher accuracies result from having more data. When one restricts to one model per trader, the model for traders with fewer transactions underperforms compared with a universal model that is trained on more datapoints. We also find that if we build a model that does not consider the inventory, cash, and recent activity of clients, that is, if the broker does not use identification of the trader, the performance of PULSE is the same. Thus, in our dataset, client-specific variables do not add value to predict the toxicity of trades.

2. Data analysis

2.1. The data

We employ data for the currency pair EUR/USD from LMAX Broker and from LMAX Exchange for the period 28 June 2022 to 21 October 2022.² For each liquidity taking trade filled by the broker, we use the direction of trade (buy or sell), the timestamp when LMAX Broker processed the trade, and the volume of the trade. Also, we use the best quotes and volumes available in LMAX Exchange at a microsecond frequency. In contrast to LMAX Broker, traders who interact in the limit order book (LOB) of LMAX Exchange do not know the identity of their counterparties. The LOB uses a price-time priority to clear supply and demand of liquidity — as in traditional electronic order books in equity markets, such as those of Nasdaq, Euronext, and the London Stock Exchange.

Table 1 shows summary statistics for the trading activity of six clients of LMAX Broker in the pair EUR/USD.

²www.lmax.com.

	Number of trades	Total volume in €100,000,000	Avg daily volume
Client 1	312,073	43.702	0.520
Client 2	56,705	3.006	0.036
Client 3	28,185	3.278	0.039
Client 4	27,743	0.456	0.005
Client 5	23,938	27.483	0.348
Client 6	13,379	5.379	0.064
Total	462,023	83.304	1.012

Table 1: Trading activity in the pair EUR/USD between clients and LMAX Broker over the period 28 June 2022 to 21 October 2022 . Volumes are reported in one hundred million euros.

Below, we work with the data of Clients 1 to 6 in Table 1 and we assume that the broker quotes her clients the best available rates in LMAX Exchange net of fees. Transaction costs in FX are around \$3 per million euros traded (see e.g., [Cartea and Sánchez-Betancourt \(2023\)](#)), so this assumption provides clients with a discount of \$3 per million euros traded when trading with the broker.

2.2. Toxicity

In this paper, a trade is toxic over a given time window if the client can unwind the trade at a profit within the time window. Instead of classifying traders as informed or uninformed, the broker assesses the probability that each trade becomes toxic within a specified time window. Not all trades sent by better informed clients will be toxic, and not all trades sent by less informed clients will be benign. Thus, our models aim to predict price movements based on current features regardless of whether the trader is informed or not. Our methods, however, include the identity of the trader, so predicting toxicity of a trade will depend, among other features, on how often the client executed toxic trades in the past.

Denote time by $t \in \mathfrak{T} = [0, T]$, where 0 is the start of the trading day and T is the end of the trading day. From this point forward, we use ‘exchange rate’ and ‘prices’ interchangeably. The best ask price and best bid price in the LOB of LMAX Exchange are denoted by $(S_t^a)_{t \in \mathfrak{T}}$ and $(S_t^b)_{t \in \mathfrak{T}}$, respectively. Let $\mathfrak{G}$ be a toxicity horizon such that $0 < \mathfrak{G} \ll T$, and let $t \in [0, T - \mathfrak{G}]$. We define the two stopping times

$$\tau_t^+ = \inf \left\{ u \in [t, T] : S_u^b > S_t^a \right\} \quad \text{and} \quad \tau_t^- = \inf \left\{ u \in [t, T] : S_t^b > S_u^a \right\},$$

with the convention that $\inf \emptyset = \infty$. The stopping time τ_t^+ is the first time after t that the best bid price is above the best ask price at time t . If $\tau_t^+ < \infty$, a buy trade executed at S_t^a becomes profitable for the client at time τ_t^+ before the end of the trading day because the client can unwind her position and collect the profit

$$S_{\tau_t^+}^b - S_t^a > 0. \quad (2.1)$$

Similarly, τ_t^- is the first time after t that the best ask price is below the best bid price at time t . If $\tau_t^- < \infty$, a sell trade executed at S_t^b is profitable for the client at time τ_t^- before the end of the trading day because the client can unwind her position and collect the profit

$$S_t^b - S_{\tau_t^-}^a > 0. \quad (2.2)$$

Definition 1 (Toxic trade). Let $\mathfrak{G} > 0$ be a toxicity horizon. A client's buy (resp. sell) filled by the broker at time t is toxic for the broker if $\tau_t^+ \leq t + \mathfrak{G}$ (resp. if $\tau_t^- \leq t + \mathfrak{G}$).

Figure 2 plots the trajectories of S_t^a and S_t^b for EUR/USD between 10:00:00 am and 10:00:10 am in LMAX Exchange on 28 June 2022. The dotted line is the best ask price and the dash-dotted line is the best bid price. The solid horizontal lines are the best ask price and the best bid price at 10:00:00 am.

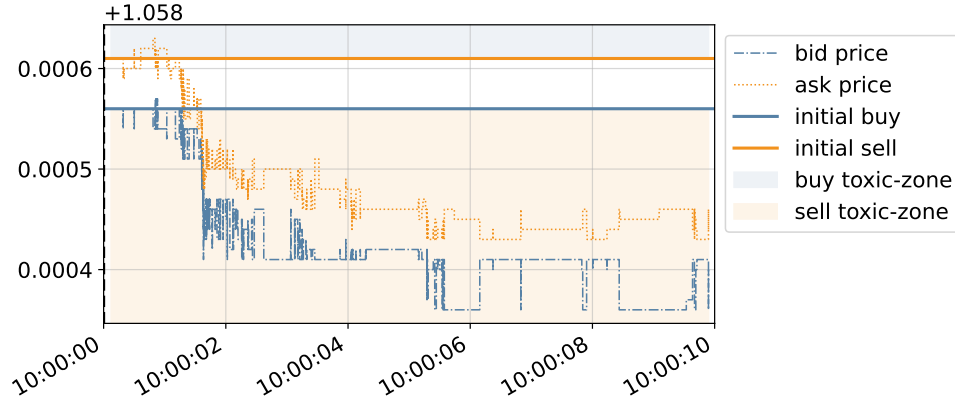


Figure 2: A sell trade from a client that becomes toxic for the broker after a few seconds of filling the trade.

In the figure, if a client buys from the broker at the best ask price at time $t = 10:00:00$ am, then there is no opportunity for the trader to unwind the trade at a profit in the first ten seconds after the trade. However, had the trader sold to the broker at the best bid price at time $t = 10:00:00$ am, then shortly after 10:00:01 the trade would be in-the-money for the client (i.e., toxic for the broker).

For a toxicity horizon $\mathfrak{G} > 0$, using both the client data and the LOB data, we determine if the trades filled by LMAX Broker were toxic over the period $\mathfrak{G}$. Table 2 shows the percentage of toxic trades executed by each client for $\mathfrak{G} \in \{1, 5, 10, 20, 30, 40, 50, 60, 70\}$ seconds.

	toxicity horizon $\mathfrak{G}$ in seconds								
	1	5	10	20	30	40	50	60	70
Client 1	6.7	25.7	38.4	51.5	58.7	63.4	66.7	69.3	71.3
Client 2	7.0	28.6	42.4	56.1	63.0	67.5	70.7	73.1	74.9
Client 3	7.0	26.0	38.6	51.1	58.2	62.6	65.8	68.2	70.4
Client 4	3.4	18.5	30.7	44.3	52.1	56.8	60.6	63.5	66.0
Client 5	8.3	22.1	31.7	42.9	50.1	55.4	59.6	62.2	63.9
Client 6	5.9	26.4	40.2	53.3	60.9	65.7	69.0	71.8	73.9
All clients	6.6	25.5	38.2	51.2	58.4	63.1	66.5	69.0	71.0

Table 2: Proportion of toxic trades (in %) between 28 June 2022 and 21 October 2022.

2.3. Features to predict toxic trades

For each client, the broker uses 247 features that reflect (i) the state of the LOB, (ii) recent activity in the LOB, and (iii) the cash and inventory of the client in the EUR/USD currency pair.³ Let $\mathcal{A} = \{c_1, \dots, c_6\}$ denote the clients of LMAX Broker in this study, and let $\mathfrak{D} := \{1, 2, \dots, 54\}$ be the 54 trading days between 28 June 2022 and 21 October 2022. For a given trading day $\mathfrak{d} \in \mathfrak{D}$, the processes

$$\left(S_t^{a,\mathfrak{d}}\right)_{t \in \mathfrak{T}}, \quad \left(S_t^{b,\mathfrak{d}}\right)_{t \in \mathfrak{T}}, \quad \left(V_t^{a,\mathfrak{d}}\right)_{t \in \mathfrak{T}}, \quad \left(V_t^{b,\mathfrak{d}}\right)_{t \in \mathfrak{T}},$$

denote the best ask price, the best bid price, the volume at the best ask price, and the volume at the best bid price in LMAX Exchange for day $\mathfrak{d}$, respectively — we drop the superscript $\mathfrak{d}$ when we do not wish to draw attention to the day. The feature associated with the log-transformed inventory of client $c \in \mathcal{A}$ is

$$\text{sign}(\mathfrak{Q}_{t-}^c) \times \log(1 + |\mathfrak{Q}_{t-}^c|),$$

where $\mathfrak{Q}_{t-}^c$ is the position in lots (one lot is €10,000) of client c accumulated over $[0, t)$ and

$$\mathfrak{Q}_t^c = \int_0^t q_u^c dN_u^{c,a} - \int_0^t q_u^c dN_u^{c,b}.$$

Here, q_t^c is the size of the order sent at time t by client c and $N_t^{c,a}$, $N_t^{c,b}$ are the counting processes of buy and sell orders, respectively, sent by client c and filled by the broker. The cash of client c , denoted by $\mathfrak{C}_t^c$, is given by

$$\mathfrak{C}_t^c = - \int_0^t S_{u-}^a q_u^c dN_u^{a,c} + \int_0^t S_{u-}^b q_u^c dN_u^{b,c},$$

and the feature associated with the cash process is

$$\text{sign}(\mathfrak{C}_{t-}^c) \times \log(1 + |\mathfrak{C}_{t-}^c|).$$

In LMAX Exchange, the bid-ask spread is

$$\mathfrak{S}_t = S_t^a - S_t^b,$$

the midprice is

$$S_t = \frac{S_t^a + S_t^b}{2},$$

the volume imbalance of the best available volumes is defined by

$$\mathfrak{I}_t = \frac{V_t^b - V_t^a}{V_t^b + V_t^a},$$

and the associated feature for the volume V is the transformed volume

$$\mathfrak{V} = \log(1 + |V|).$$

The number of trades received by the broker from her clients is

$$N_t = \sum_{c \in \mathcal{A}} \left(N_t^{c,a} + N_t^{c,b} \right),$$

³The cash and inventory of the clients are computed based on the transactions with the broker. Clients can trade elsewhere but this is unknown to the broker.

and the volatility of the midprice in the LOB of LMAX Exchange over the interval $[t - \delta, t)$ is the square root of the quadratic variation of the logarithm of the midprice over the interval. More precisely,

$$\mathfrak{V}_t^\delta = \sqrt{\sum_{\Delta \log S_u \neq 0; u \in [t-\delta, t)} |\Delta \log S_u|^2},$$

where

$$\Delta \log S_u = \log S_u - \log S_{u-}, \text{ and } S_{u-} = \lim_{v \nearrow u} S_v.$$

The return of the exchange rate of the currency pair over a period $\delta > 0$ is given by

$$\log(S_{t-}/S_{t-\delta}).$$

The timing of the events in the LOB is measured with three clocks: time-clock, transaction-clock, and volume-clock. The time-clock runs as $t \in [0, T]$ with microsecond precision, i.e., a millionth of a second. For a given day d with N^d transactions and V^d volume traded, the transaction clock runs as $\mathfrak{t} \in [0, N^d]$, and the volume-clock runs as $\mathfrak{v} \in [0, V^d]$. The number of transactions $\mathfrak{t}$ is a function of t , that is, $\mathfrak{t}(t)$ is the number of transactions observed up until time t , similar for the volume-clock $\mathfrak{v}(t)$. Thus, for any order sent at time t , the time associated with the order in the transaction-clock is $\mathfrak{t}(t)$ and the time in the volume-clock is $\mathfrak{v}(t)$.

For each clock and a given interval in the past (as measured by the clock), we employ the following eight features: (a) volatility of midprice, (b) number of trades executed by the client in the interval, (c) number of updates in the best quotes of LMAX Exchange in the interval, (d) return of midprice over the interval, (e) average transformed volume in the best bid price, (f) average transformed volume in the best ask price, (g) average spread, and (h) average imbalance of the best available volumes. Here, average refers to sample average and not clock-weighted average; our results are similar with either average. We use sample averages because it speeds up the deployment of the strategy.

More precisely, for each clock $\mathfrak{c} \in \{\text{transaction, time, volume}\}$, we build features spanning an interval $[\mathfrak{Y}_{\mathfrak{c}} 2^n, \mathfrak{Y}_{\mathfrak{c}} 2^{n+1})$ of units in the respective clock with $\mathfrak{Y}_{\mathfrak{c}} > 0$, and use a given statistic to summarise the values in the interval; for example, for the spread, the imbalance, and for the transformed volumes we use the average value over the period. In our experiments, we consider 7 intervals that span the ranges $[0, \mathfrak{Y}_{\mathfrak{c}})$ and $\{[\mathfrak{Y}_{\mathfrak{c}} 2^n, \mathfrak{Y}_{\mathfrak{c}} 2^{n+1})\}_{n=0}^8$. The median time elapsed between any two transactions for the six clients is 1.8s and the median quantity traded with LMAX Broker is €2,000. Thus, we take $\mathfrak{Y}_{\text{transaction}}$ to be one transaction, $\mathfrak{Y}_{\text{time}}$ to be one second, and we take $\mathfrak{Y}_{\text{volume}}$ to be €2,000.⁴

For a given client $c \in \mathcal{A}$, we use the following additional features: (i) cash $\mathfrak{C}_{t-}^c$, (ii) inventory $\mathfrak{Q}_{t-}^c$, (iii) volume of the order q_t^c , (iv) spread $\mathfrak{S}_{t-}$ in the market just before the order arrives, (v) imbalance $\mathfrak{I}_{t-}$ in the LOB just before the order arrives, (vi) transformed available volume in the best bid, (vii) transformed available volume in the best ask, (viii) last ask price at the time of trade, (ix) last bid price at the time of trade, (x) last midprice at the time of trade, (xi) total number of market updates since starting date, (xii) number of trades made by client c , (xiii) total number of trades made by all clients, (xiv) expanding-window volatility estimate of the mid-price, and (xv) proportion of previous sharp trades made by client c . These fifteen features account for both the

⁴The intervals for the transaction-clock are $[0, 1)$ transaction, $[1, 2)$ transactions, $[2, 4)$ transactions, ..., and $[2^5, 2^6)$ transactions. Similarly, the intervals for the time-clock are $[0, 1)$ second, $[1, 2)$ seconds, $[2, 4)$ seconds, ..., and $[2^5, 2^6)$ seconds. Lastly, the intervals in the volume-clock are $[0, \text{€}2000)$, $[\text{€}2000, \text{€}4000)$, $[\text{€}4000, \text{€}8000)$, ..., and $[\text{€}2000 \times 2^5, \text{€}2000 \times 2^6)$.

state of the LOB, and the cash and inventory of the client. The remaining 168 features account for recent activity in the LOB. These are features (a)–(h) above measured for each of the seven intervals and for each of the three clocks to obtain a total of $8 \times 7 \times 3 = 168$ features. Thus, for each client, we employ $15 + 3 \times 8 \times 7 = 183$ features to predict the toxicity of their trades.

3. Algorithm for sequential estimation of toxicity

Let $(\mathcal{F}_t)_{t \geq 0}$ be the filtration that contains the information available to the broker. For a trade (and direction of trade) from client $c \in \mathcal{A}$ filled by the broker at time t , we denote by $\mathbf{x}_t \in \mathbb{R}^M$ the features observed at time t , where M is the number of features, and the variable $y_t \in \{0, 1\}$ denotes if a period $\mathfrak{G}$ after t a trade is toxic ($y_t = 1$), where y_t is $\mathcal{F}_{t+\mathfrak{G}}$ -measurable but it is not $\mathcal{F}_t$ -measurable. We assume that the client buys or sells based on observed features $\mathbf{x}_t \in \mathbb{R}^M$ that are $\mathcal{F}_t$ -measurable. For a given trading day we create the dataset of observations $(\mathcal{D}_{t_i})_{i \in I}$, where $\mathcal{D}_t = (\mathbf{x}_t, y_t)$, $I = \{1, 2, \dots, T\}$, and T is the number of trades filled by the broker. For $i \in I$ we let t_i be the time when the order reaches the broker. To simplify notation we use $\mathcal{D}_i$ instead of $\mathcal{D}_{t_i}$; thus, we refer to the dataset $(\mathcal{D}_{t_i})_{i \in I}$ as $(\mathcal{D}_i)_{i \in I}$. For $n \in \mathbb{N}$ we let $\mathcal{D}_{1:n} = \{\mathcal{D}_1, \dots, \mathcal{D}_n\}$ be an ordered collection of observations indexed through time. Hence, for $t \leq T$ we have that $\mathcal{D}_{1:t} \subseteq \mathcal{D}_{1:T}$. We divide the dataset into a warmup dataset $\mathcal{D}_{\text{warmup}}$ and a deploy dataset $\mathcal{D}_{\text{deploy}}$. Here, $\mathcal{D}_{\text{warmup}}$ goes from 28 June 2022 to 29 July 2022, and $\mathcal{D}_{\text{deploy}}$ goes from 1 August 2022 to 21 October 2022.

We model the likelihood of a toxic trade given the features $\mathbf{x}_t$ as a Bernoulli random variable whose probability of a trade being toxic depends on the features $\mathbf{x}_t$ and a set of parameters $\boldsymbol{\theta} = (\mathbf{w}, \boldsymbol{\psi})$, that is

$$p(y | \boldsymbol{\theta}; \mathbf{x}_t) = \text{Bern}\left(y | \sigma(\mathbf{w}^\top g(\mathbf{x}_t; \boldsymbol{\psi}))\right), \quad y \in \{0, 1\}, \quad (3.1)$$

where $g : \mathbb{R}^M \rightarrow \mathbb{R}^L$ is the output-layer of a NNet. We model the last layer explicitly. Here, and throughout the paper, we adopt the convention that p denotes a likelihood function or a posterior density function. The function $\text{Bern}(\cdot | \cdot)$ is given by

$$\text{Bern}(a | b) = b^a (1 - b)^{1-a}, \quad a \in \{0, 1\} \text{ and } b \in [0, 1].$$

We refer to $\mathbf{w} \in \mathbb{R}^L$ as the last-layer parameters and we refer to $\boldsymbol{\psi} \in \mathbb{R}^D$ as the feature-transform parameters. The function $\sigma(\mathbf{w}^\top g(\mathbf{x}_t; \boldsymbol{\psi}))$ is a NNet for classification where $\sigma(x) = (1 + \exp(-x))^{-1}$ is the sigmoid function. Figure 3 shows a graphical representation of PULSE when the NNet is a multilayered-perceptron (MLP).

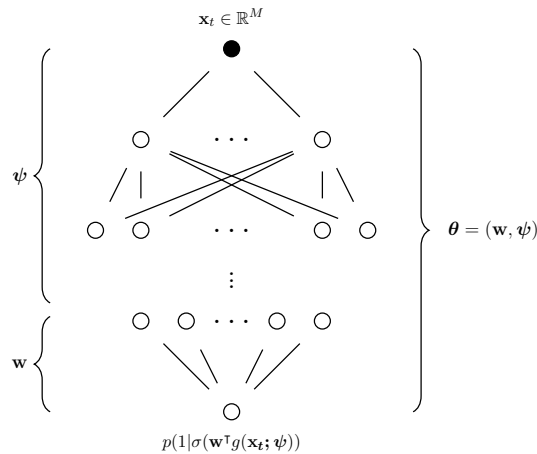


Figure 3: PULSE architecture for an MLP. The MLP is parameterised by $\boldsymbol{\theta} = (\boldsymbol{\psi}, \mathbf{w})$, where $\boldsymbol{\psi}$ are the parameters in the hidden layers and $\mathbf{w}$ are the parameters in the last layer.

Our goal is to estimate the posterior distribution for $\theta = (\mathbf{w}, \psi)$ in (3.1) after each new y_t is observed, i.e., after observing if the trade is toxic. In practice, the dimension D of the featured-transformed parameters and the dimension M of the feature space satisfy $D \gg M$, and it is costly to update θ . Larsen et al. (2021) argue that one can train a linear projection of the weights of a neural network onto a linear subspace and obtain comparable performance to that when the full set of weights is used. Furthermore, Sharma et al. (2023) show the importance of the last-layer parameters of a neural network to estimate accurately the posterior-predictive estimation of Bayesian neural networks (BNNs). Thus, to reap the benefits of last-layer methods and to have a model that can be deployed and adapted online with FX data we take two steps. One, model the posterior distribution of the last-layer parameters $\mathbf{w}$. Two, project the hidden-layer parameters ψ onto an affine subspace and model a posterior distribution of the parameters in the subspace. More precisely, we assume that $\psi \in \mathbb{R}^D$ can be decomposed as an affine projection

$$\psi = \mathbf{A} \mathbf{z} + \mathbf{b}, \quad (3.2)$$

where $\mathbf{A} \in \mathbb{R}^{D \times d}$ is the fixed projection matrix, $\mathbf{z} \in \mathbb{R}^d$ are the projected weights such that $d \ll D$, and $\mathbf{b} \in \mathbb{R}^D$ is the offset term. With this projection, we rewrite (3.1) as $p(y | \mathbf{z}, \mathbf{w}; \mathbf{x}_t) = \text{Bern}(y | \sigma(\mathbf{w}^\top g(\mathbf{A} \mathbf{z} + \mathbf{b}; \mathbf{x}_t)))$, $y \in \{0, 1\}$, and to simplify notation, we define $h(\mathbf{z}; \mathbf{x}_t) = g(\mathbf{A} \mathbf{z} + \mathbf{b}; \mathbf{x}_t)$ and write

$$p(y | \mathbf{z}, \mathbf{w}; \mathbf{x}_t) = \text{Bern}(y | \sigma(\mathbf{w}^\top h(\mathbf{z}; \mathbf{x}_t))). \quad (3.3)$$

We estimate $\mathbf{A}$, $\mathbf{b}$, and assign prior distributions for $\mathbf{w}$ and $\mathbf{z}$ in the warmup stage, and we estimate the posterior distributions for $\mathbf{w}$ and $\mathbf{z}$ in the deploy stage. Figure 4 shows these two phases, which we explain in the following two subsections.

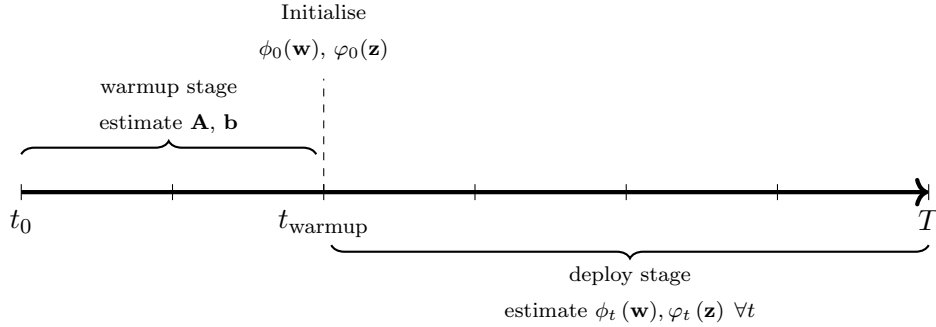


Figure 4: Warmup and deployment stages. We use all data available from times t_0 to t_{warmup} to estimate $\mathbf{A}$ and $\mathbf{b}$. At t_{warmup} , we initialise the variational approximations $\phi_0(\mathbf{w})$ and $\phi_0(\mathbf{z})$. Finally, for every $t > t_{\text{warmup}}$, we estimate $\mathbf{w}_t$ and $\mathbf{z}_t$.

3.1. Warmup phase: estimating the projection matrix and the offset term

Given the size of the dataset, we divide $\mathcal{D}_{\text{warmup}}$ into B non-intersecting random batches $\mathcal{D}_{(1)}, \dots, \mathcal{D}_{(B)}$ such that

$$\bigcup_{b=1}^B \mathcal{D}_{(b)} = \mathcal{D}_{\text{warmup}}.$$

To estimate $\mathbf{b}$ and $\mathbf{A}$, we minimise the negative loss-function

$$-\log p(\mathcal{D} | \theta) = -\sum_{n=1}^N \log p(y_t | \theta, \mathbf{x}_t) \quad (3.4)$$

via mini-batch stochastic gradient descent (SGD) over $\mathcal{D}_{\text{warmup}}$, where $\mathcal{D}$ is any random batch.

The vector $\mathbf{b}$ is given by

$$\mathbf{b} = \arg \min_{\boldsymbol{\theta}} -\log p(\mathcal{D}|\boldsymbol{\theta}). \quad (3.5)$$

The projection matrix $\mathbf{A}$ is found using singular-value decomposition (SVD) over the iterates of the SGD optimisation procedure. To avoid redundancy, we skip the first n iterations and store the iterates every k steps. The dimension of the subspace d is found via hypeparameter tuning. Convergence to a local minimum of (3.4) is obtained through multiple passes of the data. Algorithm 1 shows the training procedure for a number E of epochs.

Algorithm 1: MAP parameter estimation via batch SGD

```

1 def warmupParameters:
2   Initialise model parameters  $\boldsymbol{\theta} = (\boldsymbol{\psi}, \mathbf{w})$ 
3   foreach epoch  $e = 1, \dots, E$  do
4     foreach batch  $m = 1, \dots, M$  do
5       gradient  $= -\nabla_{\boldsymbol{\theta}} \log p(\mathcal{D}_{(m)} | \boldsymbol{\theta})$ 
6        $\boldsymbol{\theta}^{(e)} \leftarrow \boldsymbol{\theta}^{(e-1)} - \alpha \boldsymbol{\kappa}(\text{gradient})$ 
```

In Algorithm 1, the parameter α is the learning rate, and the function $\boldsymbol{\kappa} : \mathbb{R}^M \rightarrow \mathbb{R}^M$ is the per-step transformation of the Adam algorithm; see Kingma and Ba (2015). At the end of the E epochs, we obtain $\boldsymbol{\theta}^{(E)} = (\boldsymbol{\psi}^{(E)}, \mathbf{w}^{(E)})$. Then, the offset term $\mathbf{b}$ is given by

$$\mathbf{b} = \boldsymbol{\psi}^{(E)},$$

and we stack the history of the SGD iterates. To avoid redundancy, we skip the first n iterates of the SGD, which are stored at every k steps. We let

$$\mathcal{E} = \begin{bmatrix} \text{---} & \boldsymbol{\psi}^{(n)} & \text{---} \\ \text{---} & \boldsymbol{\psi}^{(n+k)} & \text{---} \\ \text{---} & \boldsymbol{\psi}^{(n+2k)} & \text{---} \\ & \vdots & \\ \text{---} & \boldsymbol{\psi}^{(E)} & \text{---} \end{bmatrix} \in \mathbb{R}^{\hat{E} \times D},$$

where $\hat{E} = E - n + 1$. With the SVD decomposition $\mathcal{E} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}$ and the first d columns of the matrix $\mathbf{V}$, the projection matrix is

$$\mathbf{A} = \begin{bmatrix} | & | & & | \\ \mathbf{V}_1 & \mathbf{V}_2 & \dots & \mathbf{V}_d \\ | & | & & | \end{bmatrix},$$

where $\mathbf{V}_k$ denotes the k -th column of $\mathbf{V}$.

3.2. Deploy phase: estimating last-layer parameters and feature-transform parameters

Given $\mathcal{D}_{\text{deploy}}$, we obtain a variational approximation to the posterior distributions of $\mathbf{w}$ and $\mathbf{z}$. We introduce Gaussian priors for both $\mathbf{w}$ and $\mathbf{z}$ at the beginning of the deploy stage. Next, let $t = 0$ denote the last timestamp in the warmup dataset and $t = 1$ the first timestamp of the deploy

dataset, and let ϕ_t and φ_t denote the posterior distribution estimates for $\mathbf{w}$ and $\mathbf{z}$ at time t . The initial estimates are given by

$$\begin{aligned}\phi_0(\mathbf{w}) &= \mathcal{N}(\mathbf{w} | \mathbf{w}^{(M)}, \sigma_{\mathbf{w}}^2 \mathbf{I}), \\ \varphi_0(\mathbf{z}) &= \mathcal{N}(\mathbf{z} | \boldsymbol{\psi}^{(M)} \mathbf{A}, \sigma_{\mathbf{z}}^2 \mathbf{I}),\end{aligned}$$

where $(\mathbf{w}^{(M)}, \boldsymbol{\psi}^{(M)})$ are the last iterates in the warmup stage, $\sigma_{\mathbf{w}}^2$ and $\sigma_{\mathbf{z}}^2$ are the coefficients of the prior covariance matrix, $\mathbf{I}$ is the identity matrix, and recall that $\mathbf{A}$ is the projection matrix.

For time $t \geq 1$, the variational posterior estimates are given by

$$\phi_t(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \boldsymbol{\nu}_t, \boldsymbol{\Sigma}_t) \quad \text{and} \quad \varphi_t(\mathbf{z}) = \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}_t, \boldsymbol{\Gamma}_t).$$

Next, to find the posterior parameters $\boldsymbol{\mu}_t, \boldsymbol{\nu}_t, \boldsymbol{\Gamma}_t, \boldsymbol{\Sigma}_t$, we recursively solve the following variational inference (VI) optimisation problem

$$\boldsymbol{\mu}_t, \boldsymbol{\nu}_t, \boldsymbol{\Gamma}_t, \boldsymbol{\Sigma}_t = \arg \min_{\boldsymbol{\mu}, \boldsymbol{\nu}, \boldsymbol{\Gamma}, \boldsymbol{\Sigma}} \text{KL}(\mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) || \phi_{t-1}(\mathbf{w}) \varphi_{t-1}(\mathbf{z}) p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)), \quad (3.6)$$

where KL is the Kullback–Leibler divergence

$$\text{KL}(p(x)||q(x)) = \int p(x) \log \left(\frac{p(x)}{q(x)} \right) dx,$$

for probability density functions p and q with the same support. The following theorem shows the update and prediction equations of the PULSE method.

Theorem 2 (PULSE). *Suppose $\log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)$ is differentiable with respect to $(\mathbf{z}, \mathbf{w})$ and the observations $\{y_t\}_{t=1}^T$ are conditionally independent over $(\mathbf{z}, \mathbf{w})$. Write the mean of the target variable y_t as a first-order approximation of the parameters centred around their previous estimate. Let $\sigma(x) = (1 + \exp(-x))^{-1}$ be the sigmoid function and $\sigma'(x) = \sigma(x)(1 - \sigma(x))$ its derivative. Then, an approximate solution to (3.6) is given by*

$$\boldsymbol{\nu}_t = \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t) \left(y_t - \sigma(\boldsymbol{\nu}_{t-1}^\top h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)) \right), \quad (3.7)$$

$$\boldsymbol{\Sigma}_t^{-1} = \boldsymbol{\Sigma}_{t-1}^{-1} + \sigma'(\boldsymbol{\nu}_{t-1}^\top h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)) h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t) h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)^\top, \quad (3.8)$$

$$\boldsymbol{\mu}_t = \boldsymbol{\mu}_{t-1} + \boldsymbol{\Gamma}_{t-1} \nabla_{\mathbf{z}} h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t) \left(y_t - \sigma(\boldsymbol{\nu}_{t-1}^\top h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)) \right), \quad (3.9)$$

$$\boldsymbol{\Gamma}_t^{-1} = \boldsymbol{\Gamma}_{t-1}^{-1} + \sigma'(\boldsymbol{\nu}_{t-1}^\top h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)) \nabla_{\mathbf{z}} h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t) \nabla_{\mathbf{z}} h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)^\top, \quad (3.10)$$

where $\boldsymbol{\mu}_t, \boldsymbol{\Gamma}_t$ are the estimated mean and covariance of the projected-hidden-layer parameters at time t and $\boldsymbol{\nu}_t, \boldsymbol{\Sigma}_t$ are the estimated mean and covariance matrix of the last-layer parameters.

See [Appendix A](#) for a proof.

4. Model evaluation

We employ the methodology developed in the previous section with the following configuration. The NNet for PULSE is an MLP with three hidden layers, 100 units in each layer, and ReLU activation function. The number of epochs E is 850, we skip the first $n = 50$ iterations of the optimisation procedure, the subspace dimension is $d = 20$, the learning rate is $\alpha = 10^{-7}$, and we store gradients every $k = 4$ steps. Under this configuration, the MLP has 38,700 units which

we estimate during the warmup stage. For the deployment stage, PULSE updates 120 degrees of freedom; this accounts for less than half a percent of all parameters updated during the warmup stage. From a practical perspective, a single step of the algorithm (during the deployment stage) incurs in a memory cost of $O((L+D)^2)$. In this paper, an update requires less than 1mb of memory because each unit is a 32bit float. Conversely, if we were not to employ the subspace approach a single step would require 190gb of memory making it infeasible to run on typical GPU devices.

4.1. Benchmarks

We compare the performance of four methods: PULSE, logistic regression (LogR), random forests (RF), and a recursively updated maximum-likelihood estimator of a Bernoulli-distributed random variable (MLE). With logistic regression, the probability that a trade is toxic is

$$p(y | \mathbf{w}_0; \mathbf{x}_t) = \text{Bern}\left(y | \sigma(\mathbf{w}_0^\top \mathbf{x}_t)\right), \quad y \in \{0, 1\}, \quad (4.1)$$

where $\mathbf{w}_0$ is estimated maximising the log-likelihood using L-BFGS; see [Liu and Nocedal \(1989\)](#).

Next, RF is a bootstrap-aggregated collection of de-correlated trees. A prediction if a trade is toxic is done by averaging over the trees, see Section 15.1 of [Hastie et al. \(2001\)](#).

Further, we model the unconditional probability of a toxic trade as a Bernoulli-distributed random variable whose mass function is given by

$$p(y | \pi) = \text{Bern}\left(y | \pi\right), \quad y \in \{0, 1\}. \quad (4.2)$$

The maximum likelihood estimator of the parameter π , given a collection $\{z_1, \dots, z_N\}$ of Bernoulli-distributed samples, where $z_n \in \{0, 1\}$ is given by

$$\pi_{\text{MLE}} = \frac{1}{N} \sum_{n=1}^N \mathbb{1}(z_n = 1);$$

here, $\mathbb{1}(\cdot)$ is the indicator function and we refer to this estimator as the MLE method. This quantity is updated after each new observation y_t .

We consider two sets of models: one for buy orders and another for sell orders trained on all clients; in 7.3, as part of the robustness analysis, we study the performance when we train one model per client ID.

5. Deployment of online methods

Next, we discuss how we evaluate the performance of the two online methods (MLE and PULSE) with asynchronous data. In classical filtering problems, as soon as new information arrives at, say, time t_i , the parameters $\boldsymbol{\theta}_{t_i}$ of the model are updated. Next, when a new trade arrives at time t_{i+1} , one uses the parameters $\boldsymbol{\theta}_{t_i}$ to estimate if a trade will be toxic. In our setting, however, an update at time t_{i+1} with $\boldsymbol{\theta}_{t_i}$ is only possible if $t_{i+1} > t_i + \mathfrak{G}$. Otherwise, we use $\boldsymbol{\theta}_{t_j}$ to predict the probability of a toxic trade, with $j = \arg \max_k t_{i+1} > t_k + \mathfrak{G}$. Figure 5 illustrates this procedure.

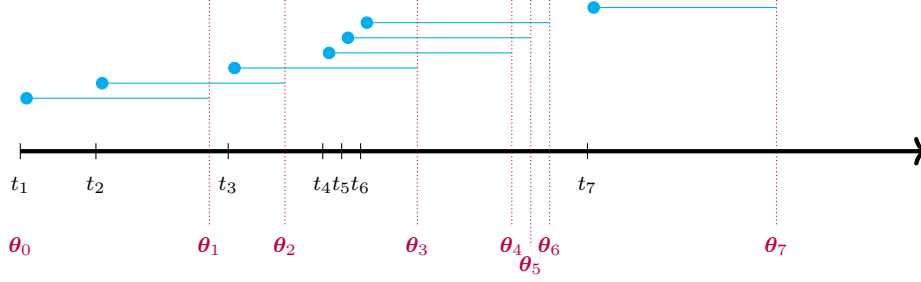


Figure 5: Asynchronous predict-update steps: trades arrive at irregular times $\{t_i\}_i$. An update to the model is only possible if $t_{i+1} > t_i + \mathfrak{S}$, when we know if the trade was toxic. In this example, the model parameters θ_0 are known at time t_1 , when a new trade arrives. When a second trade arrives, at time t_2 , we do not know if the previous trade was toxic at t_1 , so we use the model weights θ_0 to make a prediction. The next trade arrives at time $t_3 > t_1 + \mathfrak{S}$, so we use θ_1 to make a prediction. Finally, multiple trades arrive consecutively at times t_4 , t_5 , and t_6 , in which case we use θ_2 . The last trade to arrive at time t_7 uses θ_6 . In this example θ_3 , θ_4 , and θ_5 were never used to make a prediction because there are no trades between t_3 and t_6 .

We employ the asynchronous online updating for PULSE and MLE. We select hyperparameters over the warmup stage. Figure 6 shows how PULSE updates model parameters based on: current model parameters θ , features $\mathbf{x}$, and outcome y . Here, $\theta = (\psi, \mathbf{w})$ are the model parameters one uses to produce $p(y = 1 | \mathbf{x}, \theta)$, with which we compute the prediction $\hat{y}$. We employ the predictions $\hat{y}$ and the outcomes y to compute the accuracy defined above.

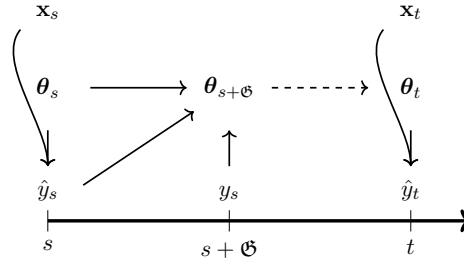


Figure 6: PULSE update procedure. For simplicity, we take $s + \mathfrak{S} < t$.

5.1. Model for decision making

Here, we devise brokerage strategies that employ predictions of toxic flow. We introduce a one-shot optimisation problem for the broker's strategy to internalise-externalise trades.

For method $M \in \{\text{PULSE}, \text{LogR}, \text{RF}, \text{MLE}\}$, let $p^{+,M} \in (0, 1)$, denote the probability that a buy order will be toxic and let $p^{-,M}$ denote the probability that a sell order will be toxic. Note that $p^{+,M} + p^{-,M}$ does not necessarily add to 1. Let $\mathfrak{S} > 0$ denote half the bid-ask spread and let $\eta > 0$ denote the shock to the midprice S if the trade is toxic; here we assume that $\eta \in (2\mathfrak{S}, \infty)$. The broker controls $\delta^\pm \in \{0, 1\}$. When $\delta^\pm = 0$ the broker externalises the trade and when $\delta^\pm = 1$ the broker internalises the trade. The broker has inventory $Q \in \mathbb{R}$; when $Q > 0$ the broker is long and when $Q < 0$ the broker is short. Assume all trades are for one unit of the asset. Then, the broker solves

$$\delta^{\pm*} = \arg \max_{\delta^\pm \in \{0,1\}} \mathbb{E} \left[\underbrace{\pm \delta^\pm (S \pm \mathfrak{S})}_{\text{cash flow}} + \underbrace{(S \pm \eta Z) (Q \mp \delta^\pm)}_{\text{inventory valuation}} - \underbrace{\phi (Q \mp \delta^\pm)^2}_{\text{inventory penalty}} \right], \quad (5.1)$$

where Z is a Bernoulli random variable with parameter $p^{\pm, M}$, and $\phi \geq 0$ is an inventory penalty parameter. Intuitively, the broker optimises the expected value of her mark-to-market adjusted by a quadratic penalty on inventory. The solutions to (5.1) are

$$\delta^{\pm*} = \mathbb{1} \left(\frac{\mathfrak{S}}{\eta} - \frac{\phi}{\eta} \pm \frac{2\phi}{\eta} Q > p^{\pm, M} \right) = \mathbb{1} (\mathfrak{p} \pm \Phi Q > p^{\pm, M}), \quad (5.2)$$

where $\mathfrak{p} := \mathfrak{S}/\eta - \phi/\eta$ and $\Phi := 2\phi/\eta$. We call Φ the inventory aversion parameter and we call $\mathfrak{p}$ the cutoff probability. The strategy internalises trades when the prediction $p^{\pm, M}$ is smaller than the cutoff probability $\mathfrak{p}$ adjusted by the inventory of the broker Q and the inventory aversion parameter Φ . When either

$$\mathfrak{p} + \Phi Q = p^{+, M} \quad \text{or} \quad \mathfrak{p} - \Phi Q = p^{-, M},$$

the broker is indifferent between internalising or externalising the trade in the market; this happens with probability zero. Below, we study the case when $\Phi = 0$ in more detail. In 7.1 we explore the case when $\Phi > 0$.

5.2. Internalise-externalise strategy

Motivated by the mathematical framework in Subsection 5.1, below we introduce a family of predictions of toxicity based on the cutoff probability $\mathfrak{p} \in [0, 1]$.

Definition 3 ($\mathfrak{p}$ -predicted toxic trade). *Let $\mathfrak{p} \in [0, 1]$ and $p(y = 1 | \mathbf{x}_{t_n}, \boldsymbol{\theta})$ be the output of a classifier. A trade is predicted to be toxic with cutoff probability $\mathfrak{p}$ if*

$$p(y = 1 | \mathbf{x}_{t_n}, \boldsymbol{\theta}) > \mathfrak{p}. \quad (5.3)$$

We store the decision of a toxic trade in the variable

$$\hat{y}_{t_n}^{\mathfrak{p}} = \mathbb{1}(p(y = 1 | \mathbf{x}_{t_n}, \boldsymbol{\theta}) > \mathfrak{p}). \quad (5.4)$$

We are interested in the predictive performance of the models as we vary $\mathfrak{p}$. To this end, let $y_{t_n} \in \{0, 1\}$ denote if a trade executed at time t_n was toxic at time $t_n + \mathfrak{S}$ ($y_{t_n} = 1$ if toxic and $y_{t_n} = 0$ otherwise). We employ the true positive rate and the false positive rate, which we define below.

Definition 4. *The true positive rate (TPR) of a sequence of trades $\{y_{t_n}\}_{n=1}^N$ with predictions $\{\hat{y}_{t_n}\}_{n=1}^N$ at a cutoff probability $\mathfrak{p}$ is*

$$TPR_{\mathfrak{p}} = \frac{\sum_{n=1}^N \mathbb{1}(y_{t_n} = \hat{y}_{t_n}^{\mathfrak{p}}) \cdot \mathbb{1}(y_{t_n} = 1)}{\sum_{n=1}^N \mathbb{1}(y_{t_n} = 1)}. \quad (5.5)$$

Definition 5. *The false positive rate (FPR) of a sequence of trades $\{y_{t_n}\}_{n=1}^N$ with predictions $\{\hat{y}_{t_n}\}_{n=1}^N$ at a cutoff probability $\mathfrak{p}$ is*

$$FPR_{\mathfrak{p}} = \frac{\sum_{n=1}^N \mathbb{1}(y_{t_n} \neq \hat{y}_{t_n}^{\mathfrak{p}}) \cdot \mathbb{1}(y_{t_n} = 0)}{\sum_{n=1}^N \mathbb{1}(y_{t_n} = 0)}. \quad (5.6)$$

Each choice of $\mathfrak{p}$ induces a pair of values $(FPR_{\mathfrak{p}}, TPR_{\mathfrak{p}})$. The graph of $\mathfrak{p} \rightarrow (FPR_{\mathfrak{p}}, TPR_{\mathfrak{p}})$ is known as the Receiver Operating Characteristic (ROC). Figure 7 shows the daily ROC of the models in the deploy stage with toxicity horizon of 30s.

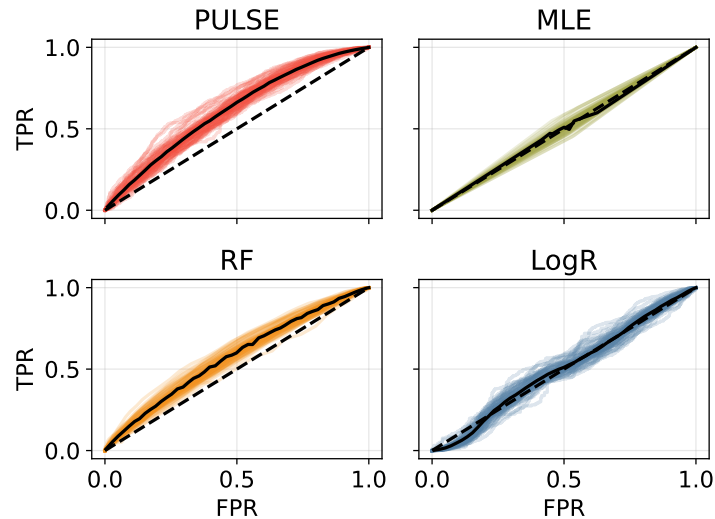


Figure 7: Daily ROC curves with toxicity horizon of 30s. We plot the daily ROC curve for each model at the end of each trading day. Each coloured line represents the ROC curve for a trading day. The solid black line is the average of the daily ROC curves. Finally, the black dashed line represents the ROC curve for a random classifier.

The area under an ROC curve, called AUC, is used in the machine learning literature to compare classifiers; see e.g., [Fawcett \(2006\)](#). Intuitively, the AUC is a measure to quantify a classifier's ability to distinguish between toxic and benign trades.

5.3. Model comparison

In this section, we analyse the AUC of the methods for varying toxicity horizons. Figure 8 shows the daily density estimate of AUC for each of the methods. That is, for each method, we compute the AUC for the sequence of trades of each day and show a density plot of such values; recall that the deploy window is between 1 August 2022 and 21 October 2022.

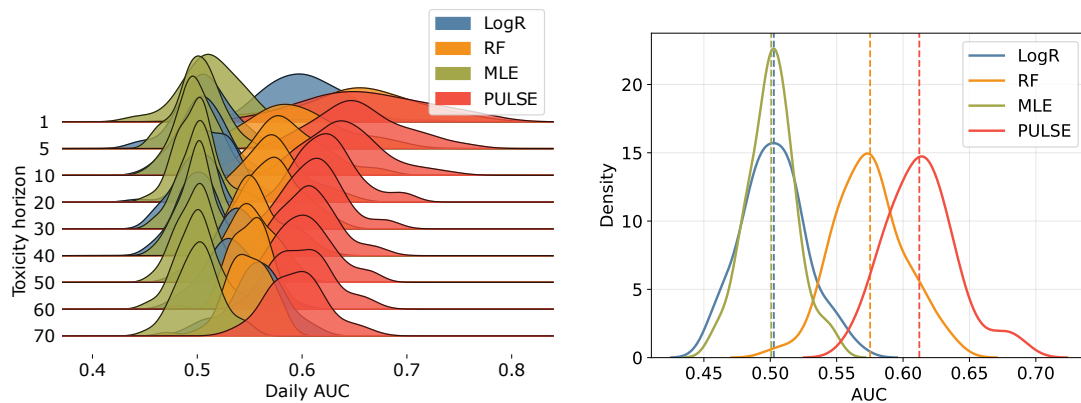


Figure 8: Left panel shows the kernel density estimate plot of daily AUC by toxic horizon. Right panel shows the daily AUC of all models for a toxicity horizon of 30s. The dashed vertical lines correspond to the mean daily value. EUR/USD currency pair over the period 1 August 2022 to 21 October 2022.

PULSE has the highest average AUC among the four methods for all toxicity horizons. Furthermore, as the toxic horizon increases, the AUC of PULSE and RF decreases, but the outperformance of PULSE over RF increases. MLE and LogR have the poorest performance.

Figure 9 shows the five-day exponentially-weighted average of the AUC for each day. We see that, whereas RF, PULSE, and MLE obtain their maximum values at the beginning of the deploy period and then decay over time, PULSE, on the other hand, maintains a steady performance. This is because PULSE updates its parameters with each new observation.

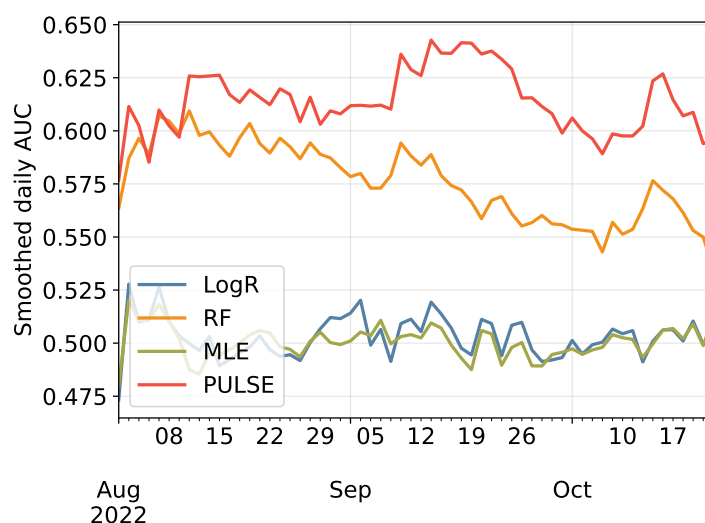


Figure 9: Five-day exponentially-weighted moving average of AUC over time. The toxicity horizon is 30 seconds.

Figure 10 shows a five-day exponentially-weighted moving average (with decay 1/3) of the AUC for the various toxicity horizons across time.

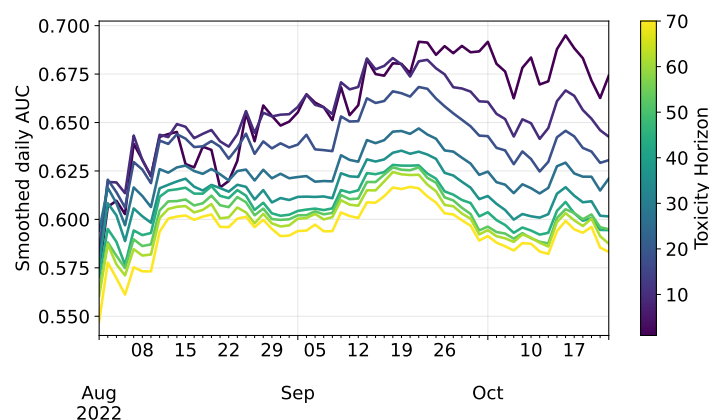


Figure 10: Five-day exponentially-weighted moving average of AUC over time for PULSE across toxicity horizons.

We see that as the toxicity horizon increases, the maximum AUC decreases. In all cases, the AUC for PULSE remains stable throughout the deploy period.

6. Trade Prediction Effectiveness and Missed Opportunities

We employ data for the EUR/USD currency pair over the period 1 August 2022 to 21 October 2022 to test the internalisation-externalisation strategy. As above, we ignore inventory aversion

(i.e., $\Phi = 0$); in Section 7 we perform robustness checks for the inventory aversion and study how the performance of the internalisation-externalisation strategy changes as the broker's inventory aversion increases.

Figure 11 reports the PnL (y -axis) of the internalised trades and the avoided PnL (x -axis) of the externalised trades when the broker uses the internalisation-externalisation strategy (5.2) with cutoff probability $\mathbf{p} \in \{0.05, 0.15, 0.25, \dots, 0.95\}$ and for PULSE, MLE, LogR, and RF. The broker starts with zero inventory at the beginning of 1 August 2022 and she crosses the spread to unwind the internalised trades at the end of the toxicity horizon, to later report the avoided PnL. We assume there is no cost incurred in externalising trades and we keep track of the PnL they would have had over the toxicity horizon. The inventory is in euros (€) and the PnL in dollars (\$). We take each trade to be for the median quantity which is €2,000 in our dataset. When a trade is toxic, the median cost to unwind the position is $-\$0.00007$ and when the trade is not toxic, the median profit is $\$0.00008$.

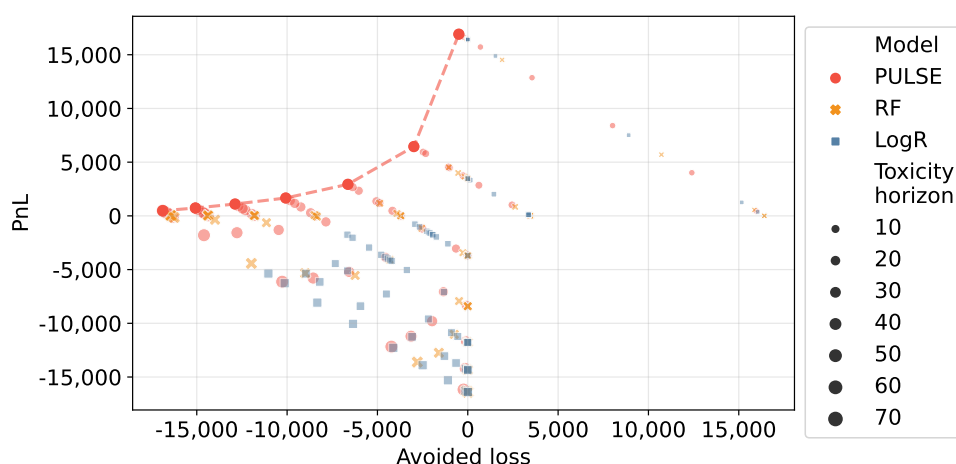


Figure 11: PnL and avoided loss for various toxicity horizons and for $\mathbf{p} \in \{0.05, 0.15, 0.25, \dots, 0.95\}$.

For each of the toxicity horizons, the internalisation-externalisation strategy informed by PULSE attains the highest PnL and the lowest avoided loss. This is indicated with the red dots joined by the red dash line. The best possible outcome is for the ten-second toxicity horizon. Next, Figure 12 shows how the percentage of internalised volume depends on the cutoff probability.

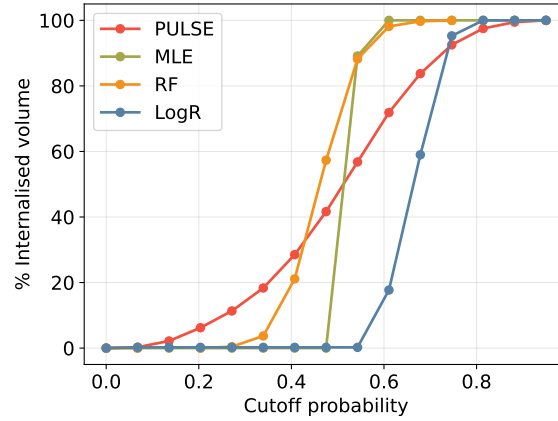


Figure 12: Percentage of internalised volume as a function of the cutoff probability p for toxicity horizon of 20s. EUR/USD currency pair over the period 1 August 2022 to 21 October 2022.

Figure 12 shows that when p takes values in the range 35% to 65%, PULSE goes from internalising around 48% of the total volume to internalising 85% of the total order flow. On the other hand, RF, LogR, and MLE exhibit abrupt changes in the percentage of internalised volume. Figure 13 shows the above curves across toxicity horizons.

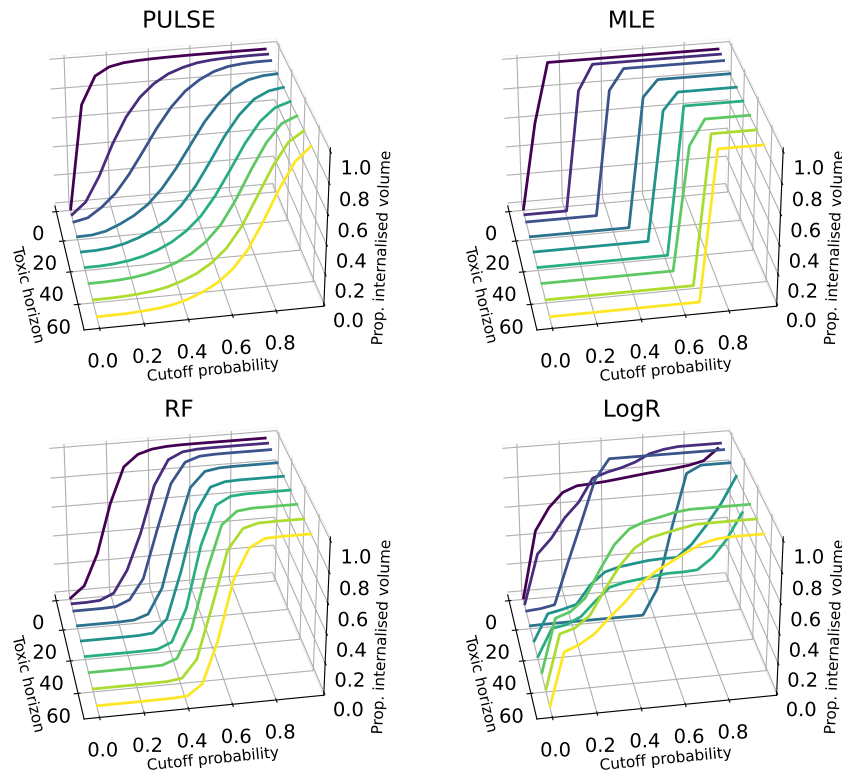


Figure 13: Proportion of internalised volume as a function of the cutoff probability p and toxicity horizon. EUR/USD currency pair over the period 1 August 2022 to 21 October 2022.

Next, Figure 14 shows a histogram of $p^{\pm, M}$ for $M \in \{\text{PULSE}, \text{MLE}, \text{LogR}, \text{RF}\}$ and toxicity

horizon of 10s.

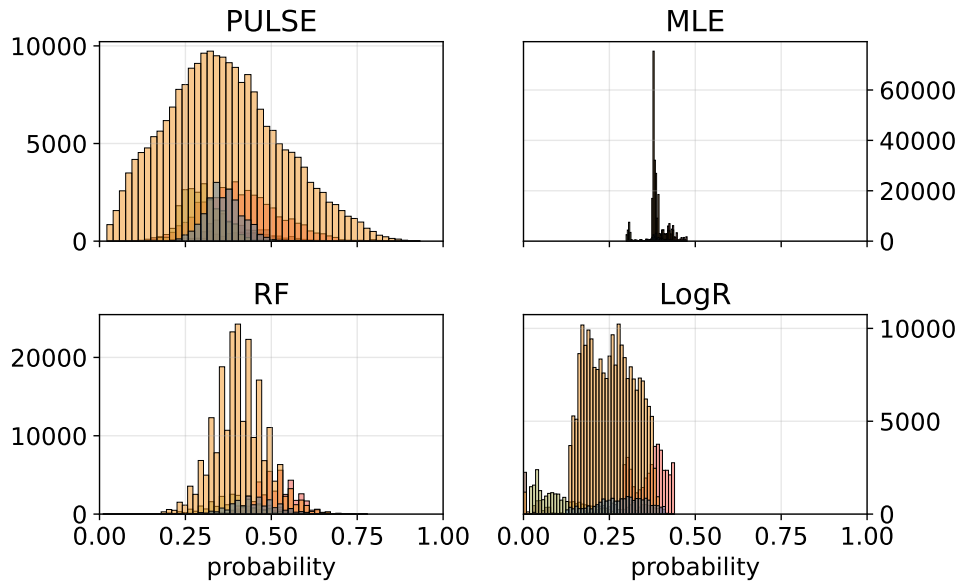


Figure 14: Histogram of predicted probability of a toxic trade. Toxicity horizon is ten seconds. EUR/USD currency pair over the period 1 August 2022 to 21 October 2022.

The predictions of PULSE are distributed sparsely. This is a desirable feature when the broker wishes to find an optimal cutoff probability to internalise-externalise trades. For example, the concentration of the predictions we see for MLE in the top right panel of Figure 14 explains the abrupt changes we see in Figure 12 and Figure 13 for MLE, LogR, and RF.

7. Robustness analysis

7.1. Inventory aversion

In this section, we study how the value of inventory aversion parameter Φ affects the results of the AUC and the internalisation-externalisation strategy. Intuitively, as the value of inventory aversion parameter increases, the broker is willing to increase the cutoff probability to internalise a trade if it would reduce the absolute value of her inventory; similarly, she is willing to decrease the cutoff probability to internalise a trade if internalising the trade increases the absolute value of her inventory.

A higher value of the inventory aversion parameter does not necessarily mean that the broker externalises more trades, it means that the broker is keener to externalise trades that decrease the absolute value of her inventory and less keen to internalise trades that increase the absolute value of her inventory. Figure 15 (left) shows the internalised volume as a function of the cutoff probability p and the inventory aversion parameter Φ . Figure 15 (right) shows the PnL and avoided loss of the internalisation-externalisation strategy in (5.2) as a function of the inventory aversion parameter Φ for a toxicity horizon of 60s using PULSE.

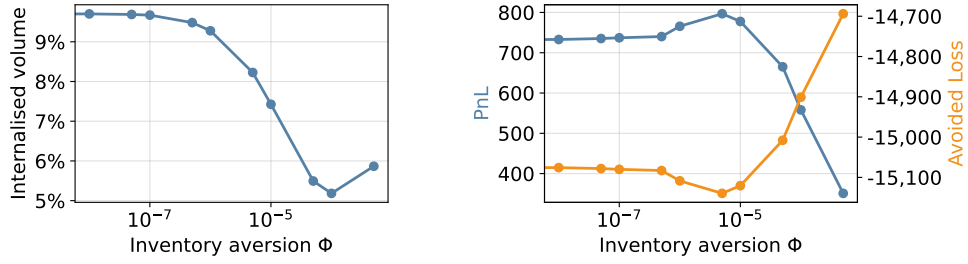


Figure 15: Left panel: percentage of internalised volume as a function of the inventory aversion parameter Φ . Right panel: PnL and avoided loss of the internalisation-externalisation strategy as a function of the inventory aversion parameter Φ .

As the value of the inventory aversion parameter Φ increases, roughly, the broker internalises less of the order flow. On the other hand, the performance of the internalisation-externalisation strategy is stable for small values of the value of the inventory aversion parameter. When the value of the inventory aversion parameter becomes large, the strategy performs poorly.

Figure 16 shows the daily volume of the internalisation-externalisation strategy in (5.2) as a function of the inventory aversion parameter Φ for a toxicity horizon of 60s using PULSE. This is a detailed version of what the left panel of Figure 15 describes. Indeed, the higher the value of the inventory aversion parameter, the less daily internalised volume.

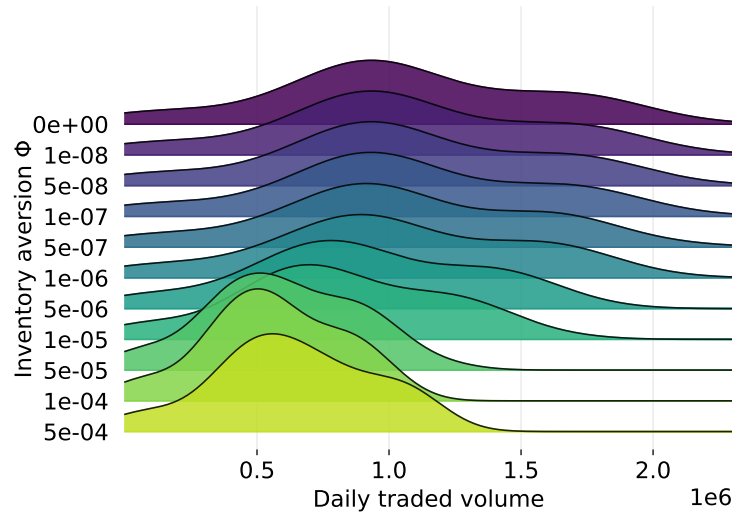


Figure 16: Kernel density estimate plot of daily traded volume for a range of values of the inventory aversion parameter Φ .

7.2. Performance for independent clocks

In Section 2.3 we construct 168 of the 175 features using three clocks: transaction-clock, time-clock, and volume-clock. Here, we explore the AUC of our models when we use only one of the three clocks to construct the features, we omit MLE because its predictions are feature-free. Recall that 168 out of the 183 features used in the calculations are measurements of 8 variables with three clocks and seven time horizons ($168 = 3 \times 8 \times 7$, see Subsection 2.3). Here, instead, we evaluate the model using 56 clock-features instead of 168, that is, we use either (i) the transaction-clock

(txn), (ii) the time-clock (time), or (iii) the volume-clock (vol). Table 3 shows the AUCs, where we observe that the models are robust to the choice of clock.

	all	time	txn	vol
PULSE	62.5	62.6	62.6	62.6
LogR	50.0	50.0	50.0	50.0
RF	56.8	56.3	53.8	53.1

Table 3: AUC for toxicity horizon of 30s by model and by clock. EUR/USD currency pair over the period 1 August 2022 to 21 October 2022.

We conclude that for PULSE, considering all three clocks does not add value. For RF on the other hand, the extra feature-engineering exercise adds value.

7.3. The added value of having one model per client

Here, we study how model predictions change when we fit a model per client as opposed to one model for all clients. Figure 17 shows the outperformance in AUC from an individual model over the universal model (one model for all clients).

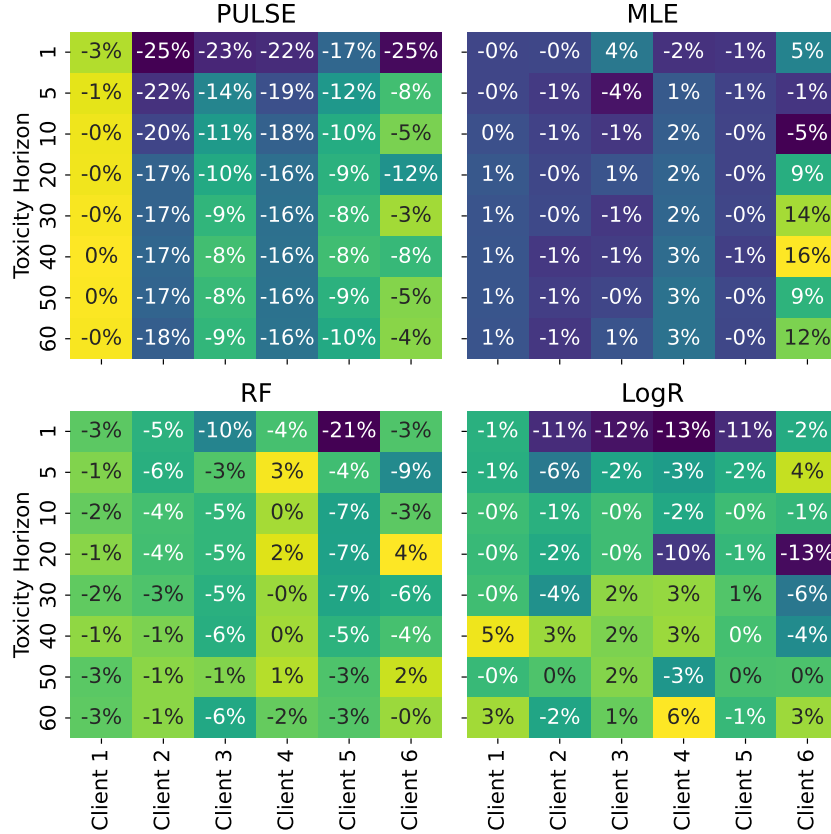


Figure 17: AUC outperformance over no-trader id

For PULSE, with the exception of Client 1, the performance of one model per client is significantly lower than that of the universal model. Indeed, the extra data is more valuable than a

model per client. The universal model employs features that are built using identity of the client, e.g., inventory and cash of client. Thus, it is more advantageous to have one model for all clients and benefit from more data points than one model per client at the expense of having fewer data points to train the individual models. This is the case for PULSE and RF, whereas the results for LogR and MLE are not conclusive. We also find that there is no statistical advantage to consider client-specific features, i.e., cash, inventory, and recent activity of clients. More precisely, accuracies remain the same when we run the models without cash, inventory, and recent activity of clients.

Lastly, we study the added value of employing data from more clients to build the models. Figure 18 shows the daily AUC distribution by toxic horizon when the models are trained with the top one hundred clients by number of trades. We do not consider client features and we use a single global model as opposed to one model per client.

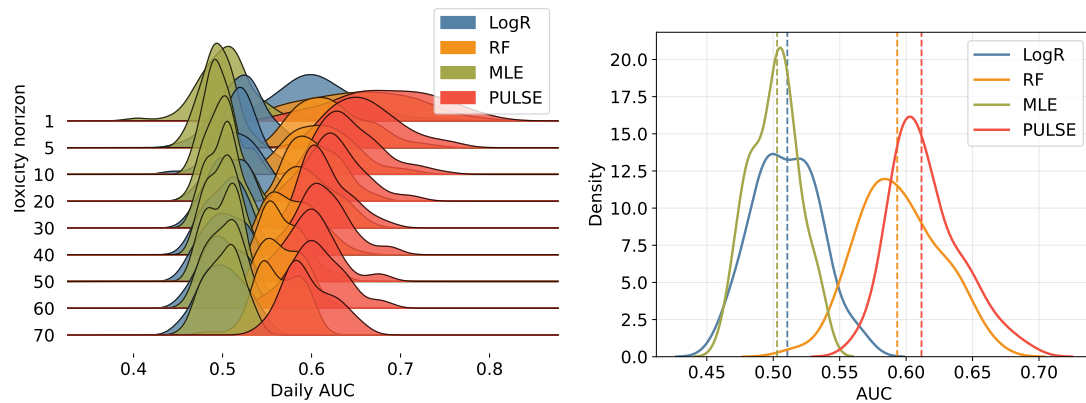


Figure 18: Left panel shows the kernel density estimate plot of daily AUC by toxic horizon. Right panel shows the daily AUC of all models for a toxicity horizon of 30s. The dashed vertical lines correspond to the mean daily value. EUR/USD currency pair over the period 1 August 2022 to 21 October 2022.

The mean daily AUC for all models performs better in Figure 18 (one global model, no features with client information, and top one hundred clients) than that in Figure 8 (one model per client, features with client information, and top six clients). In particular, the outperformance of PULSE (in AUC) ranges from 1% to 9% across toxicity horizons. Thus, in our dataset, having more clients – hence more data – is advantageous. Note that PULSE is the best model.

8. Conclusions

We employed machine learning and statistical methods to detect toxic flow. We also developed a new method, which we call PULSE, for brokers to assess the toxicity of each trade they do with their clients. Out-of-sample AUC of PULSE is high, stable, and outperforms the other methods we considered. We proposed a broker’s strategy that uses these predictions to decide which trades are internalised and which are externalised by the broker. The mean PnL of the internalise-externalise strategy we obtain with PULSE is the highest (when compared with the benchmarks) and it is robust to model parameter choices. Future research will consider a hierarchical version of the problem, where there is a structure for toxicity common to all traders. PULSE can also be used in other areas of financial mathematics, such as in the prediction of fill-rate probabilities, and in multi-armed bandit problems for trading; see, e.g., [Arroyo et al. \(2023\)](#) and [Cartea et al. \(2023\)](#).

References

- Amihud, Y. and Mendelson, H. (1980). Dealership market: Market-making with inventory. *Journal of financial economics*, 8(1):31–53.
- Arroyo, Á., Cartea, Á., Moreno-Pino, F., and Zohren, S. (2023). Deep attentive survival analysis in limit order books: Estimating fill probabilities with convolutional-transformers. *arXiv preprint arXiv:2306.05479*.
- Bagehot, W. (1971). The only game in town. *Financial Analysts Journal*, 27(2):12–14.
- Butz, M. and Oomen, R. (2019). Internalisation by electronic FX spot dealers. *Quantitative Finance*, 19(1):35–56.
- Cartea, Á., Drissi, F., and Osselin, P. (2023). Bandits for algorithmic trading with signals. *Available at SSRN 4484004*.
- Cartea, Á. and Sánchez-Betancourt, L. (2022). Brokers and informed traders: dealing with toxic flow and extracting trading signals. *Available at SSRN*.
- Cartea, Á. and Sánchez-Betancourt, L. (2023). Optimal execution with stochastic delay. *Finance and Stochastics*, 27(1):1–47.
- Copeland, T. E. and Galai, D. (1983). Information effects on the bid-ask spread. *The Journal of Finance*, 38(5):1457–1469.
- Duran-Martin, G., Kara, A., and Murphy, K. (2022). Efficient online bayesian inference for neural bandits. In *International Conference on Artificial Intelligence and Statistics*, pages 6002–6021. PMLR.
- Easley, D., Kiefer, N. M., O’Hara, M., and Paperman, J. B. (1996). Liquidity, information, and infrequently traded stocks. *The Journal of Finance*, 51(4):1405–1436.
- Fawcett, T. (2006). An introduction to roc analysis. *Pattern Recognition Letters*, 27(8):861–874. ROC Analysis in Pattern Recognition.
- Glosten, L. R. and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of financial economics*, 14(1):71–100.
- Grossman, S. J. and Stiglitz, J. E. (1980). On the impossibility of informationally efficient markets. *The American Economic Review*, 70(3):393–408.
- Hastie, T., Tibshirani, R., and Friedman, J. (2001). *The Elements of Statistical Learning*. Springer Series in Statistics. Springer New York Inc., New York, NY, USA.
- Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. In Bengio, Y. and LeCun, Y., editors, *3rd International Conference on Learning Representations, ICLR 2015*.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica: Journal of the Econometric Society*, pages 1315–1335.
- Kyle, A. S. (1989). Informed speculation with imperfect competition. *The Review of Economic Studies*, 56(3):317–355.
- Lambert, M., Bonnabel, S., and Bach, F. (2021). The recursive variational Gaussian approximation (R-VGA). *Statistics and Computing*, 32(1):10.
- Larsen, B. W., Fort, S., Becker, N., and Ganguli, S. (2021). How many degrees of freedom do we need to train deep networks: a loss landscape perspective.
- Lin, W., Khan, M. E., and Schmidt, M. (2019). Stein’s lemma for the reparameterization trick with exponential family mixtures.
- Liu, D. C. and Nocedal, J. (1989). On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45(1-3):503–528.
- Murphy, K. P. (2022). *Probabilistic Machine Learning: An Introduction*. MIT Press.
- Murphy, K. P. (2023). *Probabilistic Machine Learning: Advanced Topics*. MIT Press.
- Ollivier, Y. (2017). Online natural gradient as a Kalman filter.
- Oomen, R. (2017). Execution in an aggregator. *Quantitative Finance*, 17(3):383–404.
- Sharma, M., Farquhar, S., Nalisnick, E., and Rainforth, T. (2023). Do Bayesian neural networks need to be fully stochastic? In Ruiz, F., Dy, J., and van de Meent, J.-W., editors, *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pages 7694–7722. PMLR.

Appendix A. PULSE derivation and proofs

In this section, we present the derivation and proofs for PULSE. Proposition 1 derives the general fixed point equations. Given that these are expensive to compute, we present additional results to obtain the computationally efficient form of the theorem. We then prove Theorem 2 in Appendix A.1.

Proposition 1. (Modified R-VGA for PULSE) Suppose $\log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)$ is differentiable with respect to $(\mathbf{z}, \mathbf{w})$ and the observations $\{y_t\}_{t=1}^T$ are conditionally independent over $(\mathbf{z}, \mathbf{w})$. Given Gaussian prior distributions ϕ_0, φ_0 for $\mathbf{w}$ and $\mathbf{z}$ respectively, the variational posterior distributions at time $t \in \{1, \dots, T\}$ that solve (3.6) satisfy the fixed-point equations

$$\begin{aligned}\boldsymbol{\nu}_t &= \boldsymbol{\nu}_{t-1} - \boldsymbol{\Sigma}_{t-1} \nabla_{\boldsymbol{\nu}} \mathbb{E}_{\phi_t(\mathbf{w}), \varphi_t(\mathbf{z})} [\log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)], \\ \boldsymbol{\mu}_t &= \boldsymbol{\mu}_{t-1} - \boldsymbol{\Gamma}_{t-1} \nabla_{\boldsymbol{\mu}} \mathbb{E}_{\varphi_t(\mathbf{z}), \phi_t(\mathbf{w})} [\log p(y_t | \mathbf{z}, \boldsymbol{\psi}; \mathbf{x}_t)], \\ \boldsymbol{\Sigma}_t^{-1} &= \boldsymbol{\Sigma}_{t-1}^{-1} + 2 \nabla_{\boldsymbol{\Sigma}} \mathbb{E}_{\phi_t(\mathbf{w}), \varphi_t(\mathbf{z})} [\log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)], \\ \boldsymbol{\Gamma}_t^{-1} &= \boldsymbol{\Gamma}_{t-1}^{-1} + 2 \nabla_{\boldsymbol{\Gamma}} \mathbb{E}_{\varphi_t(\mathbf{z}), \phi_t(\mathbf{w})} [\log p(y_t | \mathbf{z}, \boldsymbol{\psi}; \mathbf{x}_t)].\end{aligned}\tag{A.1}$$

Proof. First, rewrite (3.6). Let $p(y_t) \equiv p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)$ to simplify notation. The loss function is

$$\begin{aligned}\mathcal{K}_t &= \text{KL}(\mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) || \phi_{t-1}(\mathbf{w}) \varphi_{t-1}(\mathbf{z}) p(y_t)) \\ &= \iint \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) \mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) \log \left(\frac{\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) \mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma})}{\varphi_{t-1}(\mathbf{z}) \phi_{t-1}(\mathbf{w}) p(y_t)} \right) d\mathbf{z} d\mathbf{w} \\ &= \iint \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) \mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) \left[\log \left(\frac{\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma})}{\varphi_{t-1}(\mathbf{z})} \right) + \log \left(\frac{\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma})}{\phi_{t-1}(\mathbf{w})} \right) - \log p(y_t) \right] d\mathbf{z} d\mathbf{w} \\ &= \int \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) \log \left(\frac{\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma})}{\varphi_{t-1}(\mathbf{z})} \right) d\mathbf{z} + \int \mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) \log \left(\frac{\mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma})}{\phi_{t-1}(\mathbf{w})} \right) d\mathbf{w} \\ &\quad + \iint \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) \mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) \log p(y_t) d\mathbf{w} d\mathbf{z} \\ &= \mathbb{E}_{\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma})} \left[\log \left(\frac{\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma})}{\varphi_{t-1}(\mathbf{z})} \right) \right] + \mathbb{E}_{\mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma})} \left[\log \left(\frac{\mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma})}{\phi_{t-1}(\mathbf{w})} \right) \right] \\ &\quad + \mathbb{E}_{\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) \mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma})} [\log p(y_t)] \\ &= \text{KL}(\mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) || \phi_{t-1}(\mathbf{w})) + \text{KL}(\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) || \varphi_{t-1}(\mathbf{z})) \\ &\quad + \mathbb{E}_{\mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma}) \mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma})} [\log p(y_t)].\end{aligned}\tag{A.2}$$

The first and second terms in (A.2) correspond to a Kullback–Leibler divergence between two multivariate Gaussians. The last term corresponds to the posterior-predictive marginal log-likelihood for the t -th observation. To minimise (A.2), we recall that the Kullback–Leibler divergence between two multivariate Gaussian is given by

$$\begin{aligned}\text{KL}(\mathcal{N}(\mathbf{x} | \mathbf{m}_1, \mathbf{S}_1) || \mathcal{N}(\mathbf{x} | \mathbf{m}_2, \mathbf{S}_2)) \\ = \frac{1}{2} [\text{Tr}(\mathbf{S}_2^{-1} \mathbf{S}_1) + (\mathbf{m}_2 - \mathbf{m}_1)^\top \mathbf{S}_2^{-1} (\mathbf{m}_2 - \mathbf{m}_1) - M + \log(|\mathbf{S}_2|/|\mathbf{S}_1|)] ,\end{aligned}$$

see Section 6.2.3 in [Murphy \(2022\)](#).

To simplify notation, let $\mathbb{E}_{\mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Gamma})} [\log p(y_t)] =: \mathcal{E}_t$. The derivative of $\mathcal{K}_t$ with respect to $\boldsymbol{\nu}$ is

$$\begin{aligned}\nabla_{\boldsymbol{\nu}} \mathcal{K}_t &= \nabla_{\boldsymbol{\nu}} (\text{KL}(\mathcal{N}(\mathbf{w} | \boldsymbol{\nu}, \boldsymbol{\Sigma}) || \phi_{t-1}(\mathbf{w})) + \mathcal{E}_t) \\ &= \nabla_{\boldsymbol{\nu}} \left(\frac{1}{2} \boldsymbol{\nu}^\top \boldsymbol{\Sigma}_{t-1}^{-1} \boldsymbol{\nu} - \boldsymbol{\nu}^\top \boldsymbol{\Sigma}_{t-1}^{-1} \boldsymbol{\nu}_{t-1} + \nabla_{\boldsymbol{\nu}} \mathcal{E}_t \right) \\ &= \boldsymbol{\Sigma}_{t-1}^{-1} \boldsymbol{\nu} - \boldsymbol{\Sigma}_{t-1}^{-1} \boldsymbol{\nu}_{t-1} + \nabla_{\boldsymbol{\nu}} \mathcal{E}_t \\ &= \boldsymbol{\Sigma}_{t-1}^{-1} (\boldsymbol{\nu} - \boldsymbol{\nu}_{t-1} - \boldsymbol{\Sigma}_{t-1} \nabla_{\boldsymbol{\nu}} \mathcal{E}_t).\end{aligned}\tag{A.3}$$

Set (A.3) to zero and solve for

$$\boldsymbol{\nu} = \boldsymbol{\nu}_{t-1} - \boldsymbol{\Sigma}_{t-1} \nabla_{\boldsymbol{\nu}} \mathcal{E}_t.$$

Next, we estimate the condition for $\boldsymbol{\Sigma}$. Use (A.2) to obtain

$$\begin{aligned} \nabla_{\boldsymbol{\Sigma}} \mathcal{K}_t &= \nabla_{\boldsymbol{\Sigma}} \left(-\frac{1}{2} \log |\boldsymbol{\Sigma}| + \frac{1}{2} \text{Tr}(\boldsymbol{\Sigma} \boldsymbol{\Sigma}_{t-1}^{-1}) + \mathcal{E}_t \right) \\ &= -\frac{1}{2} \boldsymbol{\Sigma}^{-1} + \frac{1}{2} \boldsymbol{\Sigma}_{t-1}^{-1} + \nabla_{\boldsymbol{\Sigma}} \mathcal{E}_t. \end{aligned} \quad (\text{A.4})$$

The fixed-point solution for (A.4) satisfies

$$\boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}_{t-1}^{-1} + 2 \nabla_{\boldsymbol{\Sigma}} \mathcal{E}_t.$$

We derive the fixed-point conditions for $\boldsymbol{\mu}$ and $\boldsymbol{\Gamma}$ similarly. $\square$

Corollary 1. *Suppose $\log p(y | \mathbf{z}, \mathbf{w})$ is differentiable with respect to $(\mathbf{z}, \mathbf{w})$ and the observations $\{y_t\}_{t=1}^T$ are conditionally independent over $(\mathbf{z}, \mathbf{w})$. Given Gaussian prior distributions ϕ_0, φ_0 for $\mathbf{w}$ and $\mathbf{z}$ respectively, the modified R-VGA equations for PULSE in order-2 form are*

$$\begin{aligned} \boldsymbol{\nu}_t &= \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} \mathbb{E}_{\phi_t(\mathbf{w}) \varphi_t(\mathbf{z})} [\nabla_{\mathbf{w}} \log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)], \\ \boldsymbol{\mu}_t &= \boldsymbol{\mu}_{t-1} + \boldsymbol{\Gamma}_{t-1} \mathbb{E}_{\phi_t(\mathbf{w}) \varphi_t(\mathbf{z})} [\nabla_{\mathbf{z}} \log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)], \\ \boldsymbol{\Sigma}_t^{-1} &= \boldsymbol{\Sigma}_{t-1}^{-1} - \mathbb{E}_{\phi_t(\mathbf{w}) \varphi_t(\mathbf{z})} [\nabla_{\mathbf{w}}^2 \log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)], \\ \boldsymbol{\Gamma}_t^{-1} &= \boldsymbol{\Gamma}_{t-1}^{-1} - \mathbb{E}_{\varphi_t(\mathbf{z})} [\nabla_{\mathbf{z}}^2 \log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)]. \end{aligned}$$

Proof. The proof follows directly by rearranging the order of integration, Bonnet's Theorem, and Prices's Lemma, see Lemma 1 and Theorem 1 in Lin et al. (2019). $\square$

Corollary 1 provides tractable fixed-point equations. Note that in Proposition 1 the gradients are taken with respect to model parameters outside the expectation, whereas in Corollary 1 the gradients and Hessians are taken inside the expectation.

Proposition 2. *Given a logistic regression model written in the canonical form*

$$\log p(y_t) = y_t \log \left(\frac{\sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)}{1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)} \right) - (1 + \exp(\sigma(\boldsymbol{\theta}^\top \mathbf{x}_t))) ,$$

we have that the gradient of $\log \left(\frac{\sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)}{1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)} \right)$ with respect to $\boldsymbol{\theta}$ is given by

$$\nabla_{\boldsymbol{\theta}} \log \left(\frac{\sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)}{1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)} \right) = \mathbf{x}_t.$$

Proof. Use the identity $\frac{d}{dx} \sigma(x) = \sigma(x) (1 - \sigma(x)) =: \sigma'(x)$ and write

$$\begin{aligned} \nabla_{\boldsymbol{\theta}} \log \left(\frac{\sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)}{1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)} \right) &= \nabla_{\boldsymbol{\theta}} \log (\sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)) - \nabla_{\boldsymbol{\theta}} \log (1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)) \\ &= (\sigma(\boldsymbol{\theta}^\top \mathbf{x}_t))^{-1} \sigma'(\boldsymbol{\theta}^\top \mathbf{x}_t) \mathbf{x}_t + (1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}_t))^{-1} \sigma'(\boldsymbol{\theta}^\top \mathbf{x}_t) \mathbf{x}_t \\ &= (1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}_t) + \sigma(\boldsymbol{\theta}^\top \mathbf{x}_t)) \mathbf{x}_t \\ &= \mathbf{x}_t. \end{aligned}$$

$\square$

Appendix A.1. Proof of Theorem 2

First rewrite the log-likelihood of the target variable y_t as a member of the exponential-family. Let $f_t(\mathbf{z}, \mathbf{w}) = \mathbf{w}^\top h(\mathbf{z}; \mathbf{x}_t)$. Then,

$$\begin{aligned} \log p(y_t) &= \log \text{Bern}(y_t | \sigma(\mathbf{w}^\top h(\mathbf{z}; \mathbf{x}_t))) \\ &= y_t \log \sigma(f_t(\mathbf{z}, \mathbf{w})) + (1 - y_t) \log(1 - \sigma(f_t(\mathbf{z}, \mathbf{w}))) \\ &= y_t \log \left(\frac{\sigma(f_t(\mathbf{z}, \mathbf{w}))}{1 - \sigma(f_t(\mathbf{z}, \mathbf{w}))} \right) + \log(1 - \sigma(f_t(\mathbf{z}, \mathbf{w}))) \\ &= y_t \log \left(\frac{\sigma(f_t(\mathbf{z}, \mathbf{w}))}{1 - \sigma(f_t(\mathbf{z}, \mathbf{w}))} \right) - \log \left(1 + \exp \left(\log \left(\frac{\sigma(f_t(\mathbf{z}, \mathbf{w}))}{1 - \sigma(f_t(\mathbf{z}, \mathbf{w}))} \right) \right) \right) \\ &= y_t \eta_t - \log(1 + \exp(\eta_t)) \\ &= y_t \eta_t - A(\eta_t), \end{aligned}$$

where $\eta_t = \log \left(\frac{\sigma(f_t(\mathbf{z}, \mathbf{w}))}{1 - \sigma(f_t(\mathbf{z}, \mathbf{w}))} \right)$ is the natural parameter and $A(\eta_t)$ the log-partition function. We perform a moment-match estimation. To do this, we follow the property of exponential-family distributions, and use the results in Proposition 2. The first and second-order derivatives of the log-partition $A(\eta_t)$ are

$$\frac{\partial}{\partial \eta_t} A(\eta_t) = \mathbb{E}[y | \eta_t] = \sigma(f_t(\mathbf{z}, \mathbf{w})), \quad (\text{A.5})$$

$$\frac{\partial^2}{\partial \eta_t^2} A(\eta_t) = \text{Cov}[y | \eta_t] = \sigma'(f_t(\mathbf{z}, \mathbf{w})). \quad (\text{A.6})$$

Then, the first order approximation $\hat{\sigma}(f_t(\mathbf{z}, \mathbf{w}))$ is

$$\hat{\sigma}(f_t(\mathbf{z}, \mathbf{w})) = \sigma(\bar{f}_{t-1}) + \sigma'(\bar{f}_{t-1})(F_{t,\mathbf{z}}^\top (\mathbf{z} - \boldsymbol{\mu}_{t-1}) + F_{t,\mathbf{w}}^\top (\mathbf{w} - \boldsymbol{\nu}_{t-1})),$$

where $\bar{f}_{t-1} = f_t(\boldsymbol{\mu}_{t-1}, \boldsymbol{\nu}_{t-1})$, $F_{t,\mathbf{z}} = \nabla_{\mathbf{z}} f_t(\mathbf{z}, \boldsymbol{\nu}_{t-1})|_{\mathbf{z}=\boldsymbol{\mu}_{t-1}}$, and $F_{t,\mathbf{w}} = \nabla_{\mathbf{w}} f_t(\boldsymbol{\mu}_{t-1}, \mathbf{w})|_{\mathbf{w}=\boldsymbol{\nu}_{t-1}} = h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)$. Next, we derive the update equations for $\boldsymbol{\nu}$ and $\boldsymbol{\Sigma}$. The moment-matched log-likelihood is given by

$$\log p(y_t) = \log \mathcal{N}(y_t | \hat{\sigma}(f_t(\mathbf{z}, \mathbf{w})), \sigma(f_t(\mathbf{z}, \mathbf{w}))(1 - \sigma(f_t(\mathbf{z}, \mathbf{w})))) \quad (\text{A.7})$$

and the gradient with respect to $\mathbf{w}$ is

$$\nabla_{\mathbf{w}} \log p(y_t) = y_t F_{t,\mathbf{w}} - \hat{\sigma}(\bar{f}_{t-1}) F_{t,\mathbf{w}} = (y_t - \hat{\sigma}(\bar{f}_{t-1})) F_{t,\mathbf{w}}.$$

Now, the Hessian of the log-model with respect to $\mathbf{w}$ is

$$\begin{aligned} \nabla_{\mathbf{w}}^2 \log p(y_t) &= \nabla_{\mathbf{w}} (y_t - \hat{\sigma}(\bar{f}_{t-1})) F_{t,\mathbf{w}} \\ &= -\sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top F_{t,\mathbf{w}}. \end{aligned}$$

The update for $\boldsymbol{\Sigma}_t$, following the order-2 form of the modified equations and replacing the expectation under $\varphi_t(\mathbf{z})$ for $\varphi_{t-1}(\mathbf{z})$, is

$$\begin{aligned} \boldsymbol{\Sigma}_t^{-1} &= \boldsymbol{\Sigma}_{t-1}^{-1} - \mathbb{E}_{\phi_t(\mathbf{w}) \varphi_{t-1}(\mathbf{z})} [\nabla_{\mathbf{w}}^2 \log p(y_t)] \\ &= \boldsymbol{\Sigma}_{t-1}^{-1} - \mathbb{E}_{\phi_t(\mathbf{w}) \varphi_{t-1}(\mathbf{z})} [-\sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top F_{t,\mathbf{w}}] \\ &= \boldsymbol{\Sigma}_{t-1}^{-1} + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top F_{t,\mathbf{w}} \\ &= \boldsymbol{\Sigma}_{t-1}^{-1} + \sigma'(\boldsymbol{\nu}_{t-1}^\top h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)) h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)^\top h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t). \end{aligned}$$

Next, the update step for $\boldsymbol{\nu}_t$ becomes

$$\begin{aligned}
\boldsymbol{\nu}_t &= \boldsymbol{\nu}_{t-1} - \boldsymbol{\Sigma}_{t-1} \mathbb{E}_{\phi_t(\mathbf{w})\varphi_{t-1}(\mathbf{z})} [\nabla_{\mathbf{w}} \log p(y_t | \mathbf{z}, \mathbf{w}; \mathbf{x}_t)] \\
&= \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} \mathbb{E}_{\phi_t(\mathbf{w})\varphi_{t-1}(\mathbf{z})} [(y_t - \hat{\sigma}(\bar{f}_{t-1})) F_{t,\mathbf{z}}] \\
&= \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} \mathbb{E}_{\phi_t(\mathbf{w})\varphi_{t-1}(\mathbf{z})} \left[(y_t - \sigma(\bar{f}_{t-1}) \right. \\
&\quad \left. + \sigma'(\bar{f}_{t-1})(F_{t,\mathbf{z}}(\mathbf{z} - \boldsymbol{\mu}_{t-1}) + F_{t,\mathbf{w}}^\top (\mathbf{w} - \boldsymbol{\nu}_{t-1}))) \right] F_{t,\mathbf{z}} \\
&= \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) - \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top (\boldsymbol{\nu}_t - \boldsymbol{\nu}_{t-1}) \right) \\
&= \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_{t-1} \right) - \sigma'(\bar{f}_{t-1}) \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_t.
\end{aligned}$$

We rewrite the last equality as

$$\boldsymbol{\nu}_t + \sigma'(\bar{f}_{t-1}) \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_t = \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_{t-1} \right),$$

which implies that

$$(\mathbf{I} + \sigma'(\bar{f}_{t-1}) \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} F_{t,\mathbf{w}}^\top) \boldsymbol{\nu}_t = \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_{t-1} \right).$$

Similarly, we have that

$$\boldsymbol{\nu}_t = (\mathbf{I} + \sigma'(\bar{f}_{t-1}) \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} F_{t,\mathbf{w}}^\top)^{-1} \left(\boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_{t-1} \right) \right),$$

where

$$\begin{aligned}
(\mathbf{I} + \sigma'(\bar{f}_{t-1}) \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} F_{t,\mathbf{w}}^\top)^{-1} &= (\boldsymbol{\Sigma}_{t-1} [\boldsymbol{\Sigma}_{t-1}^{-1} + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}} F_{t,\mathbf{w}}^\top])^{-1} \\
&= [\boldsymbol{\Sigma}_{t-1}^{-1} + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}} F_{t,\mathbf{w}}^\top]^{-1} \boldsymbol{\Sigma}_{t-1}^{-1} \\
&= \boldsymbol{\Sigma}_t \boldsymbol{\Sigma}_{t-1}^{-1}.
\end{aligned}$$

Then, it follows that

$$\begin{aligned}
\boldsymbol{\nu}_t &= \boldsymbol{\Sigma}_t \boldsymbol{\Sigma}_{t-1}^{-1} \left(\boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_{t-1} \right) \right) \\
&= \boldsymbol{\Sigma}_t \boldsymbol{\Sigma}_{t-1}^{-1} \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_t \boldsymbol{\Sigma}_{t-1}^{-1} \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_{t-1} \right) \\
&= \boldsymbol{\Sigma}_t (\boldsymbol{\Sigma}_{t-1}^{-1} + \sigma'(\bar{f}_{t-1}) F_{t,\mathbf{w}}^\top \boldsymbol{\nu}_{t-1}) \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) \right) \\
&= \boldsymbol{\Sigma}_t \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) \right) \\
&= \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} F_{t,\mathbf{w}} \left(y_t - \sigma(\bar{f}_{t-1}) \right) \\
&= \boldsymbol{\nu}_{t-1} + \boldsymbol{\Sigma}_{t-1} h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t) \left(y_t - \sigma(\boldsymbol{\nu}_{t-1}^\top h(\boldsymbol{\mu}_{t-1}; \mathbf{x}_t)) \right).
\end{aligned}$$

The updates for $\boldsymbol{\mu}_t$ and $\boldsymbol{\Gamma}_t$ are obtained similarly.